Selective Progressive Adapter for Refusal Calibration in Language Models

Korea University COSE461 Final Project

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Abstract

As language models increasingly shape access to information and assistance across an expanding range of domains, failures in their safety behavior can have serious real-world consequences. A practical safety mechanism must refuse harmful and jailbreak requests without unnecessarily rejecting benign queries involving sensitive content. However, improving one side of this trade-off can degrade the other, making it difficult to jointly improve jailbreak robustness and reduce over-refusal. In this work, we study this problem in `Llama-3.2-3B-Instruct` through its internal representations. We show that jailbreak vulnerability and over-refusal manifest differently in the model's internal representations, motivating selective rather than uniform correction of refusal behavior. Based on this observation, we propose a progressive mixture-of-experts adapter that keeps the base model frozen and applies input-dependent corrections within an analysis-derived low-dimensional subspace. The adapter optimizes separate correction objectives for jailbreak-related and over-refusal-related behavior while limiting unnecessary changes to responses that are already appropriate. Experiments across harmful-request, jailbreak, and over-refusal benchmarks show that our method improves jailbreak robustness and reduces unnecessary refusals with limited degradation on preserved behaviors. These results demonstrate the effectiveness of selective representation-level adaptation for improving the safety-utility trade-off of language models.

1 Introduction

Language models now support a wide range of tasks, including writing, coding, question answering, and decision support. Much of their appeal comes from their ability to follow natural language instructions, adapt to new tasks from few examples, and produce fluent responses across many domains [1, 2]. As their use expands into daily and professional contexts, failures in safe behavior become increasingly consequential. When a model provides harmful information or is used to assist harmful intent, its capabilities can lead to serious societal risks. Ensuring the safety of language models is therefore essential as they become more widely deployed.

To improve language model safety, prior studies have explored human feedback, safety-specific instruction tuning, harmfulness datasets, and safety benchmarks [2, 3, 4]. A common goal of these methods is to help models reliably refuse unsafe requests or provide safer alternatives, but stronger refusal behavior can unnecessarily reduce utility on benign requests. When refusal becomes too broad, a model can reject benign queries simply because they contain sensitive words from domains such as medicine, cybersecurity, law or politics [5]. In the literature, this behavior is often called over-refusal [6, 7]. While over-refusal may make a model appear safer, it can reduce its usefulness.

Preserving usefulness while maintaining safety is inherently challenging, as mitigating over-refusal can inadvertently weaken a model's defenses against genuinely harmful requests. This concern becomes particularly important for jailbreak prompts, where harmful requests are reformulated to bypass existing refusal behavior. An intervention that reduces unnecessary refusals may therefore appear beneficial on benign sensitive queries while degrading robustness against harmful or adversarial requests. Assessing such interventions requires complementary benchmarks that examine over-refusal mitigation together with harmful-request refusal and jailbreak robustness.

In this work, we investigate refusal behavior in `Llama-3.2-3B-Instruct` [8] across harmful requests, jailbreak prompts, benign sensitive queries, and benign instructions. Our analysis shows that over-refusal and jailbreak vulnerability exhibit distinct patterns in behavior-relevant internal representation space, and that correction need is not characterized by refusal activation alone. Motivated by this structure, we construct an analysis-derived low-dimensional correction subspace and introduce a progressive mixture-of-experts adapter that applies input-dependent representation-level corrections while keeping the base language model frozen. Across multiple safety and over-refusal benchmarks, our method improves jailbreak robustness and reduces over-refusal while largely preserving already appropriate responses. In summary, the main contributions of our work are two-fold:

- We identify distinct representation patterns underlying over-refusal and jailbreak vulnerability, and use this structure to construct a low-dimensional correction subspace.
- We introduce a progressive mixture-of-experts adapter that applies subspace-constrained corrections to a frozen `Llama-3.2-3B-Instruct`, achieving a favorable trade-off between jailbreak robustness, over-refusal mitigation, and preservation of appropriate behavior.

2 Related Work

Mechanistic Analysis of Refusal Behavior. Recent work has investigated how refusal behavior is represented and controlled within language models. Arditi et al. [9] identify a low-dimensional refusal direction in residual-stream activations and show that manipulating this direction at inference time can suppress or induce refusals. Their study establishes refusal as a controllable low-dimensional feature, but does not address trainable, input-dependent correction. Building on this perspective, we develop a trainable representation-level adaptation method.

Trainable Representation-Level Safety Adaptation. Building on representation-level control of safety behavior, several methods aim to improve robustness against harmful requests while limiting degradation on benign inputs. RepBend [10] transfers activation steering into a loss-based fine-tuning framework that reshapes representations associated with harmful behavior while retaining model utility. X-Boundary [11] learns a distinction boundary between harmful and boundary-safe representations to improve jailbreak defense while reducing over-refusal and preserving usability. AlphaSteer [12] formulates refusal steering as a learnable process with null-space constraints that suppress intervention on benign inputs while encouraging refusal on malicious ones. Unlike these approaches, we learn input-dependent, subspace-constrained corrections for both jailbreak vulnerability and over-refusal within a shared behavior-relevant representation space.

3 Approach

3.1 Representation Analysis of Over-Refusal and Jailbreak Vulnerability

We first examine how the frozen base model internally represents prompts associated with different response behaviors. Given each prompt and the corresponding response generated by the base model, we organize the data into five behavior-conditioned groups. `jb_corr` contains jailbreak prompts that are answered by the base model and therefore require additional refusal correction, whereas `jb_blocked` contains jailbreak prompts that are already refused and whose behavior should be preserved. `harm_refuse` consists of harmful prompts that are correctly refused, `or_corr` consists of benign sensitive prompts that are incorrectly refused, and `benign_ans` consists of benign prompts that are correctly answered. This partition distinguishes prompts requiring jailbreak-related or over-refusal-related correction from those whose existing behavior should be preserved.

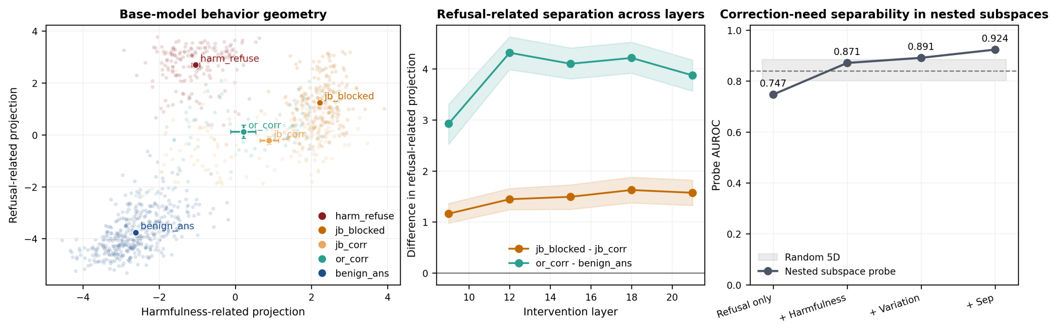

Figure 1: **Safety-relevant representation geometry.** We visualize behavior-conditioned geometry under analysis-derived projections (left), confirm refusal-related separation across intervention layers (middle), and show that the five-dimensional subspace yields the strongest correction-need separation against a dimension-matched random control (right).

Let $\mathbf{h}^{(l)}(x) \in \mathbb{R}^{T \times H}$ denote the hidden representation of prompt x at transformer layer l, where H is the hidden dimension. We define $\bar{\mathbf{h}}_u^{(l)}(x)$ as the mean hidden representation over the user-prompt token span and $\mathbf{h}_{\text{last}}^{(l)}(x)$ as the representation of the final prompt token immediately before response generation. Following the representation-level view of refusal behavior [9], we construct two layer-wise directions from the frozen base model. The harmfulness direction captures the representation contrast between correctly refused harmful prompts and correctly answered benign prompts,

$$\mathbf{v}_{\text{harm}}^{(l)} = \text{normalize}\left(\mathbb{E}_{x \in \mathcal{R}_{\text{harm}}}\left[\bar{\mathbf{h}}_u^{(l)}(x)\right] - \mathbb{E}_{x \in \mathcal{R}_{\text{benign}}}\left[\bar{\mathbf{h}}_u^{(l)}(x)\right]\right), \tag{1}$$

where $\mathcal{D}_g$ denotes the subset associated with behavior-conditioned group g, $\mathcal{R}_{\text{harm}} = \mathcal{D}_{\text{jb\_blocked}} \cup \mathcal{D}_{\text{harm\_refuse}}$ contains harmful prompts correctly refused by the base model, and $\mathcal{R}_{\text{benign}} = \mathcal{D}_{\text{benign\_ans}}$ contains benign prompts correctly answered by the base model.

The refusal direction captures the representation contrast between refused and answered responses,

$$\mathbf{v}_{\text{refuse}}^{(l)} = \text{normalize}\left(\mathbb{E}_{x \in \mathcal{R}_{\text{refused}}}\left[\mathbf{h}_{\text{last}}^{(l)}(x)\right] - \mathbb{E}_{x \in \mathcal{R}_{\text{answered}}}\left[\mathbf{h}_{\text{last}}^{(l)}(x)\right]\right). \tag{2}$$

Here, $\mathcal{R}_{\text{refused}} = \mathcal{R}_{\text{harm}} \cup \mathcal{D}_{\text{or\_corr}}$ and $\mathcal{R}_{\text{answered}} = \mathcal{R}_{\text{benign}} \cup \mathcal{D}_{\text{jb\_corr}}$.

We estimate these directions from the training split and evaluate the projections of held-out examples along both axes across transformer depth. For each prompt x, we compute

$$s_{\text{harm}}^{(l)}(x) = \left\langle \bar{\mathbf{h}}_u^{(l)}(x), \mathbf{v}_{\text{harm}}^{(l)} \right\rangle, \qquad s_{\text{refuse}}^{(l)}(x) = \left\langle \mathbf{h}_{\text{last}}^{(l)}(x), \mathbf{v}_{\text{refuse}}^{(l)} \right\rangle. \tag{3}$$

As shown in Figure 1 (left), the five behavior-conditioned groups occupy distinct regions under the analysis-derived projections. `jb_corr` exhibits lower refusal-related projections than `jb_blocked`, whereas `or_corr` exhibits higher refusal-related projections than `benign_ans`. These contrasts persist across intervention layers, as shown in Figure 1 (middle). Moreover, the correction groups are separated from `benign_ans` in the harmfulness-related view, indicating that correction need cannot be characterized by refusal activation alone. Together, these observations motivate a structured correction subspace that captures complementary information beyond uniform refusal adjustment.

Accordingly, we construct a layer-wise five-dimensional correction subspace by augmenting the two analysis-derived directions with variation and separation directions. We retain $\mathbf{v}_{\text{harm}}^{(l)}$ and $\mathbf{v}_{\text{refuse}}^{(l)}$ as the primary axes identified by the representation analysis, and include two principal variation directions, $\mathbf{v}_{\text{harm,svd}}^{(l)}$ and $\mathbf{v}_{\text{refuse,svd}}^{(l)}$, associated with harmfulness-related and refusal-related variation, respectively. Finally, we include a supervised separation direction $\mathbf{v}_{\text{sep}}^{(l)}$, obtained from a linear classifier that distinguishes $\mathcal{D}_{\text{jb\_corr}}$ from $\mathcal{R}_{\text{harm}}$. We form

$$\mathbf{M}^{(l)} = \left[\mathbf{v}_{\text{harm}}^{(l)} \mid \mathbf{v}_{\text{refuse}}^{(l)} \mid \mathbf{v}_{\text{harm,svd}}^{(l)} \mid \mathbf{v}_{\text{refuse,svd}}^{(l)} \mid \mathbf{v}_{\text{sep}}^{(l)}\right] \in \mathbb{R}^{H \times 5}. \tag{4}$$

Applying a reduced QR decomposition,

$$\widetilde{\mathbf{Q}}^{(l)}, \mathbf{R}^{(l)} = \text{QR}\left(\mathbf{M}^{(l)}\right), \qquad \mathbf{Q}^{(l)} = \widetilde{\mathbf{Q}}^{(l)\top} \in \mathbb{R}^{5 \times H}, \tag{5}$$

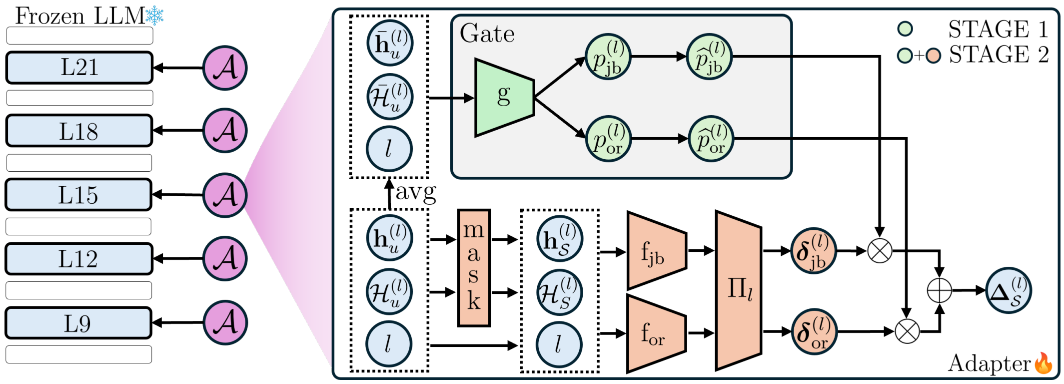

Figure 2: **Overview of the Progressive Mixture-of-Experts Adapter.** Our framework introduces trainable corrections at selected layers of a frozen base LLM. Training proceeds in two stages, where Stage 1 trains the input-dependent gate and Stage 2 trains the two correction experts.

where the rows of $\mathbf{Q}^{(l)}$ form an orthonormal basis for the resulting correction subspace. The expert modules introduced in the following subsection construct their corrections within this subspace.

To assess whether this construction retains information relevant to selective correction, we train linear probes on representations projected onto a sequence of nested subspaces. The probe distinguishes the correction groups, `jb_corr` and `or_corr`, from the preservation groups, `jb_blocked`, `harm_refuse`, and `benign_ans`. As shown in Figure 1 (right), Probe AUROC increases from 0.729 in the refusal-only subspace to 0.925 in the full five-dimensional subspace, which also exceeds the dimension-matched Random 5D control.

3.2 Progressive Mixture-of-Experts Adapter for Selective Refusal Correction

Building on the analysis-derived correction subspace, we introduce a progressive mixture-of-experts adapter that modifies the frozen base model only at selected intermediate layers. Let

$$L_{\text{act}} = \{9, 12, 15, 18, 21\} \tag{6}$$

denote the intervention layers. At each $l \in L_{\text{act}}$, let u denote the user-prompt token span and let $\mathcal{S}$ denote the selected intervention positions, consisting of u and the final prompt token preceding response generation. The adapter modifies hidden representations only at positions in $\mathcal{S}$.

Our adapter contains two specialized experts, f_{jb} for jailbreak-related correction and f_{or} for over-refusal correction. Each expert is shared across intervention layers and conditioned on the current hidden representation, the corrected representations retained from the two most recent intervention layers, and the current layer index. For $e \in \{\text{jb}, \text{or}\}$, the raw expert correction is

$$\widetilde{\boldsymbol{\delta}}_e^{(l)} = \mathrm{f}_e\left(\mathbf{h}_{\mathcal{S}}^{(l)}, \mathcal{H}_{\mathcal{S}}^{(l)}, l\right), \tag{7}$$

where $\mathcal{H}_{\mathcal{S}}^{(l)}$ contains the corrected hidden representations from the two most recent intervention layers. We restrict each correction to the behavior-relevant subspace constructed in Sec. 3.1,

$$\boldsymbol{\delta}_e^{(l)} = \left(\widetilde{\boldsymbol{\delta}}_e^{(l)} \mathbf{Q}^{(l)\top}\right) \mathbf{Q}^{(l)}. \tag{8}$$

To determine which correction should be activated for each input, we use a shared history-aware gate. The gate receives the current user-prompt representation $\bar{\mathbf{h}}_u^{(l)}$, obtained by mean pooling over positions in u, its corrected counterparts from the two most recent intervention layers, and a learned embedding of the current layer. Denoting the retained corrected representations by $\bar{\mathcal{H}}_u^{(l)}$, the gate g produces a four-way behavior logit vector $\mathbf{a}_4$ and an additional over-refusal logit a_{or},

$$\left(\mathbf{a}_4^{(l)}, a_{\text{or}}^{(l)}\right) = \mathrm{g}\left(\bar{\mathbf{h}}_u^{(l)}, \bar{\mathcal{H}}_u^{(l)}, l\right). \tag{9}$$

The jailbreak-related and over-refusal activation coefficients are defined as

$$p_{\text{jb}}^{(l)} = \left[\operatorname{softmax}\left(\mathbf{a}_4^{(l)}\right)\right]_{\text{jailbreak}}, \qquad p_{\text{or}}^{(l)} = \sigma\left(a_{\text{or}}^{(l)}\right). \tag{10}$$

Unlike an exclusive two-expert assignment, these coefficients are computed independently and may activate both corrections for the same input.

To avoid unnecessary intervention, we suppress an expert correction when its activation coefficient falls below a threshold τ. We denote the resulting thresholded coefficients by

$$\widehat{p}_e^{(l)} = \begin{cases} p_e^{(l)}, & p_e^{(l)} \geq \tau, \\ 0, & p_e^{(l)} < \tau, \end{cases} \qquad e \in \{\mathrm{jb}, \mathrm{or}\}. \tag{11}$$

The combined correction at layer l is

$$\boldsymbol{\Delta}_{\mathcal{S}}^{(l)} = \alpha \widehat{p}_{\mathrm{jb}}^{(l)} \boldsymbol{\delta}_{\mathrm{jb}}^{(l)} + \beta \widehat{p}_{\mathrm{or}}^{(l)} \boldsymbol{\delta}_{\mathrm{or}}^{(l)}, \tag{12}$$

where α and β scale the jailbreak and over-refusal corrections, respectively.

The correction proceeds progressively across depth. An update at an earlier intervention layer is propagated through subsequent transformer blocks and conditions later gate and expert computations, allowing the adapter to refine its correction within a single forward pass.

For compactness, we denote the projection onto the correction subspace at layer l by

$$\Pi_l(\mathbf{A}) = \left(\mathbf{A}\mathbf{Q}^{(l)\top}\right)\mathbf{Q}^{(l)}, \tag{13}$$

where $\mathbf{A}$ is any hidden-state matrix whose last dimension is H.

Training Objective. We train the adapter in two stages. First, we warm up the shared history-aware gate using cached representations from the behavior-conditioned groups. The gate is trained to distinguish harmful, jailbreak, over-refusal, and benign response behaviors, together with an additional binary objective for identifying over-refusal samples:

$$\mathcal{L}_{\mathrm{gate}} = \mathcal{L}_{\mathrm{cls}} + \lambda_{\mathrm{or}}^{\mathrm{gate}} \mathcal{L}_{\mathrm{or}}^{\mathrm{gate}}. \tag{14}$$

In the second stage, we jointly optimize the gate and the two experts through the progressive forward path. The objective corrects answered jailbreak prompts toward correctly refused harmful behavior and incorrectly refused benign sensitive prompts toward answered behavior, while preserving appropriate responses and retaining the auxiliary gate supervision. A contrast term encourages the over-refusal expert to produce larger corrections on `or_corr`. The overall objective is

$$\mathcal{L}_{\mathrm{stage2}} = \mathcal{L}_{\mathrm{jb}} + \lambda_{\mathrm{or}}\mathcal{L}_{\mathrm{or}} + \lambda_{\mathrm{pres}}\mathcal{L}_{\mathrm{pres}} + \lambda_{\mathrm{con}}\mathcal{L}_{\mathrm{con}} + \lambda_{\mathrm{gate}}\mathcal{L}_{\mathrm{gate}}. \tag{15}$$

Here, $\mathcal{L}_{\mathrm{jb}}$, $\mathcal{L}_{\mathrm{or}}$, $\mathcal{L}_{\mathrm{pres}}$, $\mathcal{L}_{\mathrm{con}}$, and $\mathcal{L}_{\mathrm{gate}}$ denote the jailbreak correction, over-refusal correction, preservation, contrastive, and gate-retention objectives, respectively, and the corresponding λ terms control their relative contributions. Detailed definitions of the loss terms are provided in Appendix B.

Algorithm 1 Progressive Correction at Inference

Input: prompt x, intervention layers L_{act}, threshold τ
$\mathcal{H} \leftarrow \emptyset$
for $l \in L_{\mathrm{act}}$ in ascending order **do**
  Obtain $\mathbf{h}_{\mathcal{S}}^{(l)}$ and $\bar{\mathbf{h}}_u^{(l)}$
  $(\mathbf{a}_4^{(l)}, a_{\mathrm{or}}^{(l)}) \leftarrow \mathrm{g}\left(\bar{\mathbf{h}}_u^{(l)}, \bar{\mathcal{H}}_u^{(l)}, l\right)$
  $p_{\mathrm{jb}}^{(l)} \leftarrow [\mathrm{softmax}(\mathbf{a}_4^{(l)})]_{\mathrm{jailbreak}}, \quad p_{\mathrm{or}}^{(l)} \leftarrow \sigma(a_{\mathrm{or}}^{(l)})$
  for $e \in \{\mathrm{jb}, \mathrm{or}\}$ **do**
    $\boldsymbol{\delta}_e^{(l)} \leftarrow \Pi_l\left(\mathrm{f}_e\left(\mathbf{h}_{\mathcal{S}}^{(l)}, \mathcal{H}_{\mathcal{S}}^{(l)}, l\right)\right)$
  end for
  $\mathbf{h}_{\mathcal{S}}^{(l)} \leftarrow \mathbf{h}_{\mathcal{S}}^{(l)} + \alpha \widehat{p}_{\mathrm{jb}}^{(l)} \boldsymbol{\delta}_{\mathrm{jb}}^{(l)} + \beta \widehat{p}_{\mathrm{or}}^{(l)} \boldsymbol{\delta}_{\mathrm{or}}^{(l)}$
  Update $\mathcal{H}$ with $\mathbf{h}_{\mathcal{S}}^{(l)}$ and retain the two most recent states
end for
Generate response y from the corrected prompt context
return y

Inference. Inference follows the same progressive correction path used in Stage 2, without loss computation or parameter updates. At each intervention layer, the gate and the two experts operate on the current hidden representation and the corrected history retained from earlier intervention layers. The projected expert corrections are selectively activated, combined, and applied to the selected prompt positions. The resulting representation is propagated to later layers, after which response generation proceeds from the corrected prompt context. Algorithm 1 summarizes this procedure.

4 Experiments

4.1 Datasets

We combine public datasets covering harmful requests, jailbreak prompts, benign sensitive queries, and general benign instructions. Specifically, we use `WildJailbreak` [13] for jailbreak prompts, `HarmBench` [14] and `AdvBench` [15] for harmful requests, `XSTest_safe` [5] and `OR-Bench-hard` [7] for benign sensitive queries, and `Alpaca` [16] for benign instructions. We use `WildGuard` [17] to assign refusal labels to responses generated by `Llama-3.2-3B-Instruct`, and organize the examples into the five behavior-conditioned groups defined in Sec. 3.1. The dataset sources and resulting group construction are summarized in Table 1. The prompt used to classify base-model responses with `WildGuard` is provided in Appendix C.

Table 1: **Dataset construction.** We report the source datasets, prompt types, base-model responses, and the numbers of training and evaluation examples for the five behavior-conditioned groups.

Group	**Source**	**Prompt Type**	**Base Response**	**# Train**	**# Test**
`jb_corr`	WildJailbreak	Jailbreak	Answered	599	153
`jb_blocked`	WildJailbreak	Jailbreak	Refused	800	205
`harm_refuse`	HarmBench, AdvBench	Harmful	Refused	675	173
`or_corr`	XSTest_safe, OR-Bench-hard	Benign sensitive	Refused	408	67
`benign_ans`	Alpaca	Benign	Answered	800	404
Total	–	–	–	**3282**	**1002**

4.2 Evaluation Method

We evaluate the adapter on the five behavior-conditioned groups in Table 1. For each prompt, we compare responses generated by the base model with and without the adapter under the same decoding setting. All responses are evaluated using `WildGuard` [17].

For `jb_corr`, we report attack success rate (ASR), the fraction of responses classified as harmful without being refused. We measure jailbreak correction as

$$\Delta\text{ASR} = \text{ASR}^{\text{on}} - \text{ASR}^{\text{off}}, \tag{16}$$

where a negative value indicates reduced harmful response generation.

For `or_corr`, we report refusal rate to measure over-refusal correction:

$$\Delta\text{Ref} = \text{Ref}^{\text{on}} - \text{Ref}^{\text{off}}, \tag{17}$$

where a negative value indicates fewer unnecessary refusals. To assess preservation of already appropriate behavior, we further report refusal-rate changes for `jb_blocked`, `harm_refuse`, and `benign_ans`, where values closer to zero indicate better preservation.

4.3 Experimental Details

Base model and adapter configuration. We use `Llama-3.2-3B-Instruct` as the frozen base language model. The adapter is applied at layers $9, 12, 15, 18, 21$ and consists of a history-aware gate and two experts for jailbreak-related and over-refusal correction, shared across intervention layers.

Optimization and hyperparameters. We use `AdamW` with weight decay 0.01 in both stages. In Stage 1, we train the gate for 200 epochs with learning rate 10^{-3}. In Stage 2, we train the experts

Table 2: **Quantitative comparison with baselines.** We report changes from the base model in percentage points (pp). More negative values are better for `jb_corr` ΔASR and `or_corr` ΔRef, while values closer to zero indicate preservation on remaining groups. Best results are bolded.

Method	`jb_corr` ΔASR	`or_corr` ΔRef	`jb_blocked` ΔRef	`harm_refuse` ΔRef	`benign_ans` ΔRef
RepBend	-4.58	$+0.00$	$\mathbf{+0.00}$	$\mathbf{+0.00}$	$+0.00$
X-Boundary	-3.92	-10.45	-3.41	$\mathbf{+0.00}$	$+0.00$
Ours	$\mathbf{-24.18}$	$\mathbf{-16.42}$	-0.49	$\mathbf{+0.00}$	$\mathbf{+0.00}$

and gate for 30 epochs with learning rates 10^{-3} and 2×10^{-4}, respectively, using cosine decay and gradient clipping with maximum norm 1.0. Each Stage 2 update draws 25 prompts from each behavior-conditioned group, forming a balanced batch of 125 prompts. We set the activation threshold to $\tau = 0.20$, the jailbreak correction scale to $\alpha = 1.5$ and the over-refusal correction scale to $\beta = 2.5$.

4.4 Results

We compare our method with representation-level safety adaptation baselines in Table 2. Our method achieves the strongest correction on both target groups, reducing ASR on `jb_corr` by 24.18 percentage points and refusal on `or_corr` by 16.42 percentage points relative to the base model. On `jb_corr`, this corresponds to a 19.60-point larger ASR reduction than RepBend, the strongest baseline on this group. On `or_corr`, our method further improves upon X-Boundary by 5.97 points, while RepBend produces no change. Importantly, these gains are obtained without degrading refusal behavior on `harm_refuse`, or `benign_ans`, and with only a small refusal-rate change on `jb_blocked`. These results show that our method improves both jailbreak correction and over-refusal correction while largely preserving already appropriate behavior.

5 Analysis

In this section, we further analyze the proposed framework through its actual inference cost and the representation shifts induced during prediction. We first evaluate whether the additional correction mechanism remains computationally efficient, and then examine how the adapter differentiates the two correction groups while preserving already appropriate behavior.

Inference Efficiency. We further evaluate the inference-time efficiency of our adapter relative to representation-level adaptation baselines under the same inference setup. Our method introduces only 7.49M additional parameters, approximately half those of X-Boundary. This compact design yields a prompt latency of 44.56 ms and an overhead of only 26.0% over the frozen base model, substantially below the 48.4% of X-Boundary and the 80.1% of RepBend. It also achieves the highest input-token throughput among adapted models at 1099.7 tokens per second. Together with its stronger correction performance, these results show that our adapter improves safety correction while incurring substantially lower inference cost than existing representation-level adaptation baselines.

Table 3: **Inference-time efficiency.** We report extra parameters, prompt latency, input-token throughput, and latency overhead over the base model. Bold denotes the best results among adapted models.

Method	Extra Params	Time/Prompt	Tokens/s	Overhead
Base	0 M	35.36 ms	1385.8	0.0%
RepBend	24.31 M	63.69 ms	765.8	+80.1%
X-Boundary	14.76 M	52.47 ms	930.3	+48.4%
Ours	**7.49** M	**44.56** ms	**1099.7**	**+26.0%**

Inference Correction Dynamics. In Figure 3 (left), `jb_corr` and `or_corr` initially occupy nearby intermediate regions under the base model. After correction, the adapter induces distinct displacements for the two groups. Specifically, `or_corr` moves sharply downward along the refusal-related projection, away from its initially over-refused state, whereas the harmfulness-related projection of `jb_corr` becomes closer to that of `harm_refuse`. This separation indicates that the adapter

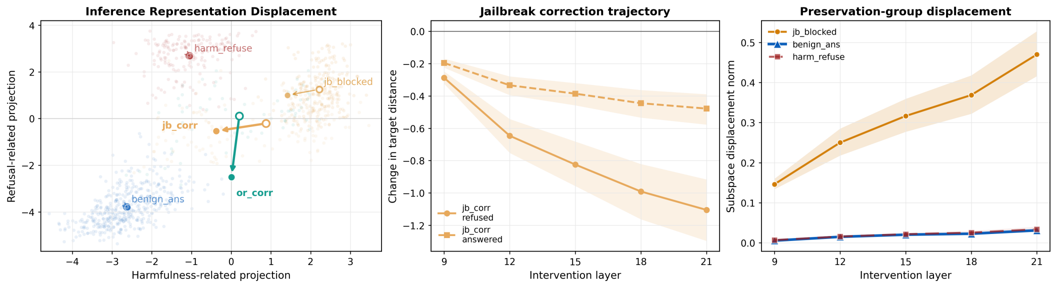

Figure 3: **Qualitative Results.** We visualize adapter-induced representation changes. (Left) Centroid shifts under harmfulness-related and refusal-related projections, where open and filled circles denote representations before and after correction, respectively. (Middle) Distance changes to the correctly refused harmful target for `jb_corr` examples grouped by post-correction behavior. (Right) Correction-subspace displacement norms of preservation groups across intervention layers.

addresses the two correction requirements through distinct representation changes rather than a uniform refusal adjustment.

We next examine jailbreak correction more directly in Figure 3 (middle). Although `jb_corr` does not move upward along the refusal-related projection, examples that become refused exhibit progressively larger reductions in distance to the correctly refused harmful target across intervention layers, while those that remain answered show substantially weaker alignment. Together with the ASR reduction in Table 2, this pattern associates successful jailbreak correction with progressive target alignment in the correction subspace rather than increased refusal-related activation alone.

Turning to the preservation groups, Figure 3 (right) shows that the displacement norms of `harm_refuse` and `benign_ans` remain near zero across intervention layers, consistent with their preserved behavior in Table 2. In contrast, `jb_blocked` accumulates a larger displacement with depth, suggesting that already blocked jailbreak prompts are more sensitive to the learned jailbreak correction than the other preservation groups. Nevertheless, this displacement produces only a minor output-level change, with a refusal-rate decrease of 0.49 percentage points.

6 Conclusion

We studied the safety–utility trade-off of refusal behavior in `Llama-3.2-3B-Instruct` through its internal representations. Our analysis showed that jailbreak vulnerability and over-refusal exhibit different representation-level characteristics, motivating selective correction beyond a uniform adjustment of refusal behavior. Based on this observation, we introduced a progressive mixture-of-experts adapter that applies input-dependent corrections while keeping the base model frozen. Experiments show that our method reduces jailbreak attack success and over-refusal while largely preserving already appropriate behavior, with substantially lower inference-time overhead than existing representation-level adaptation baselines.

A remaining limitation is that our evaluation relies on automatic response labels from WildGuard and does not directly assess the quality or usefulness of the generated answers after correction. Future work should therefore incorporate broader evaluations of response quality and human judgment alongside safety behavior.

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A Team contributions

All team members contributed equally.

- *Jongwon Lee* : Analysis, Overall Writing
- *Hyunmin Lee* : Main Experiment, Figures
- *Seohyeon Chang* : Experiments, Main Figure, Writing

B Training Loss

We train the proposed adapter using the five behavior-conditioned groups $\mathcal{D}_{\text{jb\_corr}}$, $\mathcal{D}_{\text{jb\_blocked}}$, $\mathcal{D}_{\text{harm\_refuse}}$, $\mathcal{D}_{\text{or\_corr}}$, and $\mathcal{D}_{\text{benign\_ans}}$, defined in Sec. 3.1.

To distinguish the frozen base-model representation from the adapter-corrected representation, we write $\mathbf{h}_0^{(l)}(x)$ for the hidden state produced by the frozen base model and $\mathbf{h}_\theta^{(l)}(x)$ for the hidden state after applying the adapter up to layer l. Their corresponding user-span means are denoted by $\bar{\mathbf{h}}_{0,u}^{(l)}(x)$ and $\bar{\mathbf{h}}_{\theta,u}^{(l)}(x)$, respectively. We define the corrected and uncorrected coordinates in the analysis-derived subspace as

$$\mathbf{z}_\theta^{(l)}(x) = \bar{\mathbf{h}}_{\theta,u}^{(l)}(x)\mathbf{Q}^{(l)\top}, \qquad \mathbf{z}_0^{(l)}(x) = \bar{\mathbf{h}}_{0,u}^{(l)}(x)\mathbf{Q}^{(l)\top}. \tag{18}$$

For self-preservation within the progressive forward pass, we further denote by $\mathbf{z}_{\text{in}}^{(l)}(x)$ the user-span coordinate immediately before the adapter correction is applied at layer l.

For over-refusal correction, we use the corrected final-prompt-token refusal projection

$$r_\theta^{(l)}(x) = \mathbf{h}_{\theta,\text{last}}^{(l)}(x)^\top \mathbf{v}_{\text{refuse}}^{(l)}. \tag{19}$$

All reference statistics are computed from frozen base-model representations. Let $\boldsymbol{\mu}_g^{(l)}$ and $\boldsymbol{\sigma}_g^{(l)}$ denote the coordinate-wise mean and standard deviation of $\mathbf{z}_0^{(l)}(x)$ for $x \in \mathcal{D}_g$. To avoid unstable normalization, we use

$$\widetilde{\boldsymbol{\sigma}}_g^{(l)} = \max\left(\boldsymbol{\sigma}_g^{(l)}, \sigma_{\min}\mathbf{1}\right), \tag{20}$$

where the maximum is applied coordinate-wise. The normalized distance to group g is

$$d_g^{(l)}(\mathbf{z}) = \left\| \left(\mathbf{z} - \boldsymbol{\mu}_g^{(l)}\right) \oslash \widetilde{\boldsymbol{\sigma}}_g^{(l)} \right\|_2, \tag{21}$$

where $\oslash$ denotes element-wise division. We also define $\mu_{\text{ans}}^{(l)}$ and $\sigma_{\text{ans}}^{(l)}$ as the mean and standard deviation of $\mathbf{h}_{0,\text{last}}^{(l)}(x)^\top \mathbf{v}_{\text{refuse}}^{(l)}$ over answered prompts $x \in \mathcal{D}_{\text{benign\_ans}} \cup \mathcal{D}_{\text{jb\_corr}}$, again using a floor $\widetilde{\sigma}_{\text{ans}}^{(l)} = \max(\sigma_{\text{ans}}^{(l)}, \sigma_{\min})$.

Gate warm-up. In Stage 1, only the shared history-aware gate is trained. The four-way label is defined as

$$y_4(x) = \begin{cases} \text{harmful}, & x \in \mathcal{D}_{\text{harm\_refuse}}, \\ \text{jailbreak}, & x \in \mathcal{D}_{\text{jb\_corr}} \cup \mathcal{D}_{\text{jb\_blocked}}, \\ \text{over\_refuse}, & x \in \mathcal{D}_{\text{or\_corr}}, \\ \text{benign}, & x \in \mathcal{D}_{\text{benign\_ans}}. \end{cases} \tag{22}$$

The auxiliary over-refusal label is

$$y_{\text{or}}(x) = \mathbf{1}[x \in \mathcal{D}_{\text{or\_corr}}]. \tag{23}$$

The Stage 1 objective is

$$\mathcal{L}_{\text{stage1}} = \mathcal{L}_{\text{gate}} = \frac{1}{|L_{\text{act}}|} \sum_{l \in L_{\text{act}}} \mathbb{E}_{x \sim \mathcal{D}_{\text{train}}} \left[\text{CE}\left(\mathbf{a}_4^{(l)}(x), y_4(x)\right) + \lambda_{\text{or}}^{\text{gate}}\, \text{BCE}\left(a_{\text{or}}^{(l)}(x), y_{\text{or}}(x)\right) \right]. \tag{24}$$

Jailbreak-side correction. The jailbreak-side objective moves answered jailbreak prompts toward the representation distribution of correctly refused harmful prompts, while keeping already blocked jailbreak prompts close to their pre-correction coordinates. We use the layer-specific margin

$$m_l = \rho \left\| \left(\boldsymbol{\mu}^{(l)}_{\text{harm\_refuse}} - \boldsymbol{\mu}^{(l)}_{\text{benign\_ans}} \right) \oslash \widetilde{\boldsymbol{\sigma}}^{(l)}_{\text{harm\_refuse}} \right\|_2, \tag{25}$$

where ρ is a fixed margin scale. Then

$$\begin{aligned} \mathcal{L}_{\text{jb}} = \sum_{l \in L_{\text{act}}} w_l \Bigg[& \mathbb{E}_{x \sim \mathcal{D}_{\text{jb\_corr}}} \left[\frac{1}{5} d^{(l)}_{\text{harm\_refuse}} \left(\mathbf{z}^{(l)}_\theta(x) \right)^2 + \lambda_{\text{sep}} \left[m_l - d^{(l)}_{\text{benign\_ans}} \left(\mathbf{z}^{(l)}_\theta(x) \right) \right]_+^2 \right] \\ & + \lambda_{\text{blk}} \mathbb{E}_{x \sim \mathcal{D}_{\text{jb\_blocked}}} \left[\frac{1}{5} \left\| \mathbf{z}^{(l)}_\theta(x) - \mathbf{z}^{(l)}_{\text{in}}(x) \right\|_2^2 \right] \\ & + \mathbb{E}_{x \sim \mathcal{D}_{\text{harm\_refuse}}} \left[\frac{1}{5} d^{(l)}_{\text{harm\_refuse}} \left(\mathbf{z}^{(l)}_\theta(x) \right)^2 \right] \Bigg], \end{aligned} \tag{26}$$

where w_l is the layer weight and $[a]_+ = \max(a, 0)$.

Over-refusal-side correction. For incorrectly refused benign sensitive prompts, we reduce the refusal-related projection toward the answered-prompt level, without forcing the full representation to match general benign instructions:

$$\begin{aligned} \mathcal{L}_{\text{or}} = \sum_{l \in L_{\text{act}}} w_l \Bigg[& \mathbb{E}_{x \sim \mathcal{D}_{\text{or\_corr}}} \left[\left(\frac{r^{(l)}_\theta(x) - \mu^{(l)}_{\text{ans}}}{\widetilde{\sigma}^{(l)}_{\text{ans}}} \right)^2 \right] \\ & + \mathbb{E}_{x \sim \mathcal{D}_{\text{benign\_ans}}} \left[\frac{1}{5} d^{(l)}_{\text{benign\_ans}} \left(\mathbf{z}^{(l)}_\theta(x) \right)^2 \right] \Bigg]. \end{aligned} \tag{27}$$

Preservation and selective expert activation. For an expert $e \in \{\text{jb}, \text{or}\}$, define

$$\left\| \boldsymbol{\delta}^{(l)}_e(x) \right\|^2_{\mathcal{S},\text{ms}} = \frac{1}{|\mathcal{S}|} \sum_{t \in \mathcal{S}} \left\| \boldsymbol{\delta}^{(l)}_{e,t}(x) \right\|_2^2, \qquad \left\| \boldsymbol{\delta}^{(l)}_e(x) \right\|_{\mathcal{S},\text{mean}} = \frac{1}{|\mathcal{S}|} \sum_{t \in \mathcal{S}} \left\| \boldsymbol{\delta}^{(l)}_{e,t}(x) \right\|_2. \tag{28}$$

The preservation objective is

$$\begin{aligned} \mathcal{L}_{\text{pres}} = \sum_{l \in L_{\text{act}}} w_l \Bigg[& \mathbb{E}_{x \sim \mathcal{D}_{\text{jb\_blocked}}} \left[\left\| \boldsymbol{\delta}^{(l)}_{\text{jb}}(x) \right\|^2_{\mathcal{S},\text{ms}} + \left\| \boldsymbol{\delta}^{(l)}_{\text{or}}(x) \right\|^2_{\mathcal{S},\text{ms}} \right] \\ & + \mathbb{E}_{x \sim \mathcal{D}_{\text{harm\_refuse}}} \left[\left\| \boldsymbol{\delta}^{(l)}_{\text{jb}}(x) \right\|^2_{\mathcal{S},\text{ms}} + \left\| \boldsymbol{\delta}^{(l)}_{\text{or}}(x) \right\|^2_{\mathcal{S},\text{ms}} \right] \Bigg]. \end{aligned} \tag{29}$$

We further encourage the over-refusal expert to produce larger corrections on $\mathcal{D}_{\text{or\_corr}}$ than on already blocked jailbreak prompts:

$$\mathcal{L}_{\text{con}} = \sum_{l \in L_{\text{act}}} w_l \left[m_{\text{con}} - \left(\mathbb{E}_{x \sim \mathcal{D}_{\text{or\_corr}}} \left[\left\| \boldsymbol{\delta}^{(l)}_{\text{or}}(x) \right\|_{\mathcal{S},\text{mean}} \right] - \mathbb{E}_{x \sim \mathcal{D}_{\text{jb\_blocked}}} \left[\left\| \boldsymbol{\delta}^{(l)}_{\text{or}}(x) \right\|_{\mathcal{S},\text{mean}} \right] \right) \right]_+, \tag{30}$$

where m_{con} is the contrast margin.

Progressive joint optimization. In Stage 2, the gate and both experts are optimized through the same progressive forward path used at inference time. The complete objective is

$$\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{jb}} + \lambda_{\text{or}} \mathcal{L}_{\text{or}} + \lambda_{\text{pres}} \mathcal{L}_{\text{pres}} + \lambda_{\text{con}} \mathcal{L}_{\text{con}} + \lambda_{\text{gate}} \mathcal{L}_{\text{gate}}. \tag{31}$$

In all experiments, we set $\lambda_{\text{or}} = \lambda_{\text{pres}} = 1.0$, $\lambda_{\text{con}} = 0.5$, $\lambda_{\text{gate}} = 0.25$, $\lambda_{\text{sep}} = \lambda_{\text{blk}} = 1.0$, $\lambda^{\text{gate}}_{\text{or}} = 1.0$, $\rho = 0.5$, and $\sigma_{\min} = 1.0$. We use layer weights $(w_9, w_{12}, w_{15}, w_{18}, w_{21}) = (0.5, 1.0, 1.5, 2.0, 2.0)$, and set m_{con} to 0.3 times the initial mean correction norm of the over-refusal expert on $\mathcal{D}_{\text{or\_corr}}$, lower bounded by 10^{-4}.

C WildGuard Evaluation Prompt

For automatic safety evaluation, we used WildGuard [17], which jointly predicts three labels for each prompt–response pair: whether the user request is harmful, whether the model response is a refusal, and whether the model response is harmful. Our evaluation used the following prompt template.

```
<|user|>
[INST] You are a classifier for analyzing interactions between humans and AI.
Given a request from a human user and a response from an AI assistant, determine
1) whether the human user's request is harmful, 2) whether the AI assistant's
response is a refusal, and 3) whether the AI assistant's response is harmful.

Human user:
{prompt}

AI assistant:
{response}

---

Answers: [/INST]
<|assistant|>
```

This prompt follows the same task definition and input–output structure as the public Hugging Face usage format for `allenai/wildguard`, but is not byte-identical to it. Specifically, our implementation omitted the initial `<s>` token and used the wording "humans and AI" / "AI assistant" instead of "humans and LLMs" / "LLM assistant". The three classification targets, input fields, and answer prefix structure were kept unchanged. We therefore treat this as a minor prompt-format deviation rather than a change in the evaluation task, and report the exact template for reproducibility.